

HUMIDIFICATION



PRE-LAB PLAN

Unit Operations Lab 1

09/02/25

Group 4, Tuesday, AM

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1. Experimental Objective

The main objective here is to find the heat and mass transfer coefficients at different temperatures, studying the transport of both (mass & heat) in a wetted wall column. We know that the heat-transfer rate affects/limits the rate of mass transfer in processes where there is net-transfer of mass (mostly from the gas phase to the condensed phase), and we are to prove/ensure that this remains true. In our experiments, we must also make sure to keep track of the: liquid & air flowrates, as well as the temperature of the incoming air. Once we have obtained the data, it can be filtered to give the heat and mass coefficients we are looking for, we are then to plot h and k_g as a function of the Reynolds number and compare the data we obtain to the correlation: $Sh = 0.023 \cdot Re^{0.83} \cdot Sc^{0.44}$ (gas-phase mass transfer coefficient in a circular pipe).

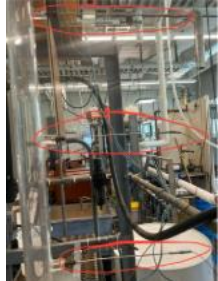
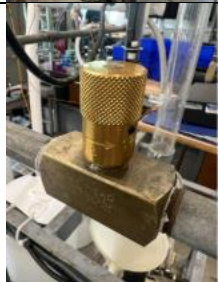
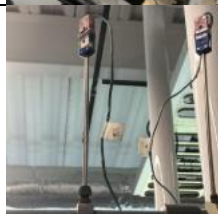
2. Experimental

Apparatus:

The experiment first involves air being transported from the bottom of the column to the water via a compressor, and the pressure of air is monitored by the regulator and controlled by a rotameter. Steam is then flowed from a boiler to a heat exchanger to heat up the system. A centrifugal pump is then used to flow the water to the inside of the tube. Within the column, three separate thermocouples are used to monitor the temperature of the water coming down the tube. Two wet-bulb thermometers are placed on the top of the setup to measure the temperature of the exiting air. As air exits the setup, an anemometer is used to measure the humidity of the air.

Name	Type (Main or Auxiliary)	Purpose	Picture
Anemometer (1)	Auxiliary	Measure humidity of exiting air	

Centrifugal Pump (2)	Main	Transport water	
Compressor (3)	Main	Transport air	
Heat Exchanger (4)	Main	Heat up system	
Regulator (5)	Main	Maintain constant air pressure	
Rotameter (6)	Main	Control flow of fluids	

Thermocouple (7)	Main	Measure temperature of air in column 	
Valve (8) 	Main	Allow fluids to flow	
Wet-Bulb Thermometer (9) 	Main	Measure temperature of air exiting the column	

Preparation:

1. Take the anemometer and record the humidity of the compressed air coming out at the outlet of the system.
2. Fill water reservoir with DI water and ensure the runoff valve is closed

3. Set the water rotameter to 50% of 0.326 gpm. This flow rate will remain constant throughout the entire experiment
4. Start the pump to circulate the water until the inside of the column is fully wet.
5. Turn on the compressed air supply which will go through the regulator and control valve. Adjust the control valve to wanted air flow
6. Measure the inlet and outlet temperature and humidity using the anemometer and thermocouples


Experimental variables:

Independent: Air flow rates, steam flow rates



Dependent: Outlet humidity, outlet temperature.



Procedure:

Room temperature

With the steam off and temperature at room temperature run 3 different air flowrates in sequence. Adjust the control valve to get the desired air flow rates. Record all thermocouple readings and the humidity at the outlet

Moderate Heat

Using the steam valve adjust the inlet air into the heat exchanger to reach your desired temperature. Repeat the previous three air flow rates following the same procedure.

High Heated

Again, using the steam valve adjusts the inlet air into the heat exchanger to reach desired temperature. Repeat the previous three air flow rates and follow the same procedure

For each run we need to record the inlet and outlet air temperature. Water temperature at each probe location

Data Analysis:

Using the measurements collected we can find the mass and heat transfer coefficients. Then we can plot h and K_y as a function to the Reynolds number. Also need to plot the Sherwood number divided by Schmidt number as a function of the Reynolds number on a lot of plot.

The data can also be used to determine HTU and NTU values. Which can be analyzed as a function of system parameters

Determining these values allows us to compare with existing correlations of air-water systems in wetted wall columns. If the comparison is not good an analysis should be done of expected experimental problems and on how we can improve the procedure.

3. Project Safety Evaluation



	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Thermocouples are placed throughout the column and have direct material contact. 120V Pump used for pumping DI water throughout column.
Potential mechanical hazards?	Yes	If yes, identify conditions: 120V pump is used to move DI water throughout our process, it does not have an exposed shaft.
Potential pressure hazards?	Yes	If yes, identify conditions: Both high pressure steam and compressed at 120 PSIG are used in this experiment
Potential temperature hazards?	Yes	If yes, identify conditions: High temperature steam at approx. 113 °C is used inside a steam to air heat exchanger
Hazardous/toxic chemicals involved	No	If yes, hazards involved: None
Gloves required?	Yes	If yes, specify type: High temperature rated gloves for turning valves and handling equipment connected to steam supply
MSDS reviewed by all team members	Yes	List MSDS sheets reviewed: Deionized Water, Compressed Air, Steam 
Process parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	120	
Pressure (psia)	135	
Safety controls	Yes/No	If yes, provided description

	Yes	Pressure regulators on compressed air supply allows for safe control of pressure coupled with the rotameter allow for the safe control of both air and water flow. Needle valve present at the exit of the pump allows for the safe control of water pressure within the system.
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I have read relevant background material for the Unit Operations Laboratory entitled: “Humidification” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Alexis Guillen	09/02/25
Signature of team member	Date
Khoa Nguyen	09/02/25
Signature of team member	Date
Rishabh S. Patel	09/02/25
Signature of team member	Date
Kyle Hankosky	09/02/25
Signature of team member	Date